

A Simple But Versatile MUSICAL HORN

S.C. DWIVEDI

This simple musical horn can be used in a car, scooter, cycle, or motorbike, or even as a call-bell. It uses melody generator IC UM66 that generates a musical tone, which is amplified by IC LM386. The tone of the horn cannot, however, be changed as

no potentiometer has been used to keep the circuit simple. The author's prototype is shown in Fig. 1.

Circuit diagram of the musical horn is shown in Fig. 2. The circuit comprises a bridge rectifier (BR1), 9V voltage regulator 7809 (IC1), melody generator UM66 (IC2), low-

power audio amplifier LM386 (IC3), and a few other components. Capacitors connected across the supply terminals are used to minimise any noise signals.

Heart of the circuit is melody generator IC UM66. The IC has an inbuilt beat and tone generator. It has three legs like a transistor and is available in many versions that produce different melodies. The UM66 series of ICs is meant for use in call bells, cell phones, and toys.

IC UM66 has a built-in ROM program for music and consumes very low power. The melody signal is generated at pin 3 of UM66 and it is amplified by LM386 to drive the loudspeaker. The circuit can generate music even without audio amplifier LM386, but the volume will be very low.

The UM66TXX series is designed specifically to generate melody for tone generation circuits. Last two digits of the IC specify the tone/melody generated by that IC. The list of some common tone generator ICs and their equivalent numbers are:

UM66T01 – Jingle bells, Santa is coming to town, and We wish you a merry Xmas

UM66T02 – Jingle bells

UM66T04 – Jingle bells, Rudolph the red-nosed reindeer, and Joy to the world

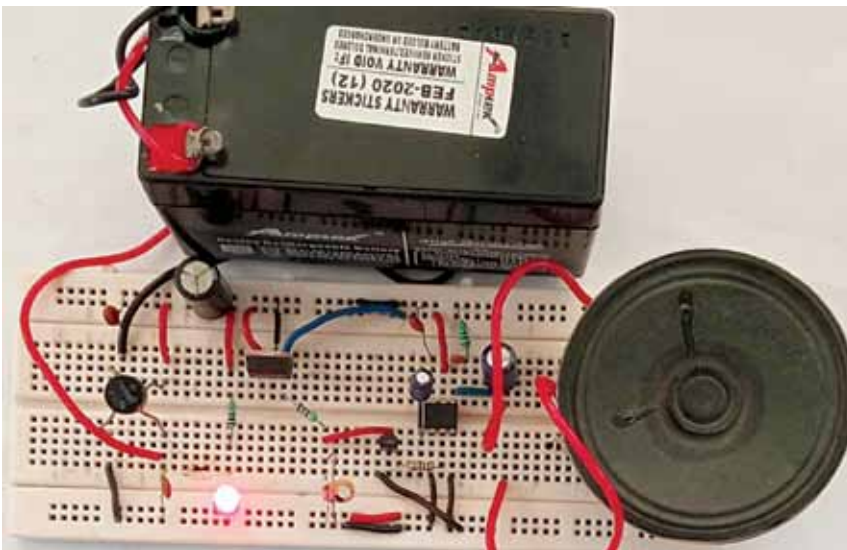


Fig. 1: Author's prototype on a breadboard

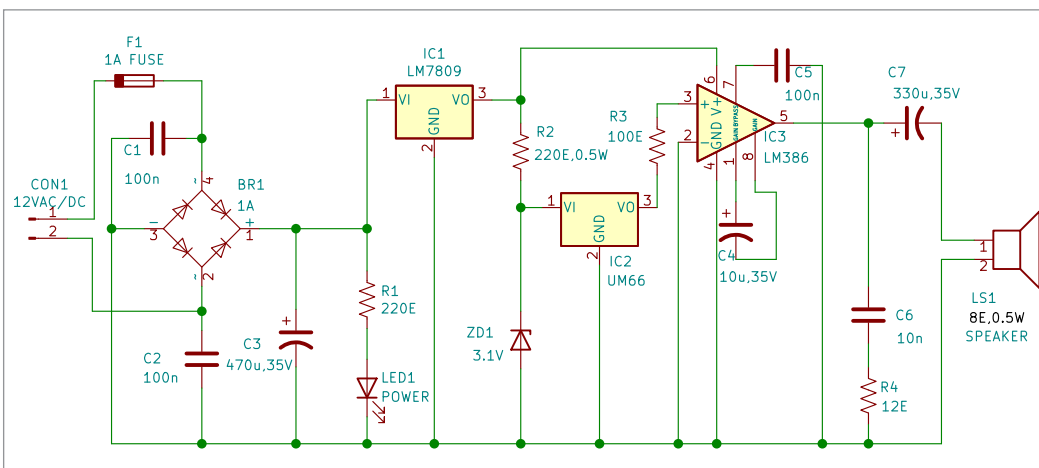


Fig. 2: Circuit diagram